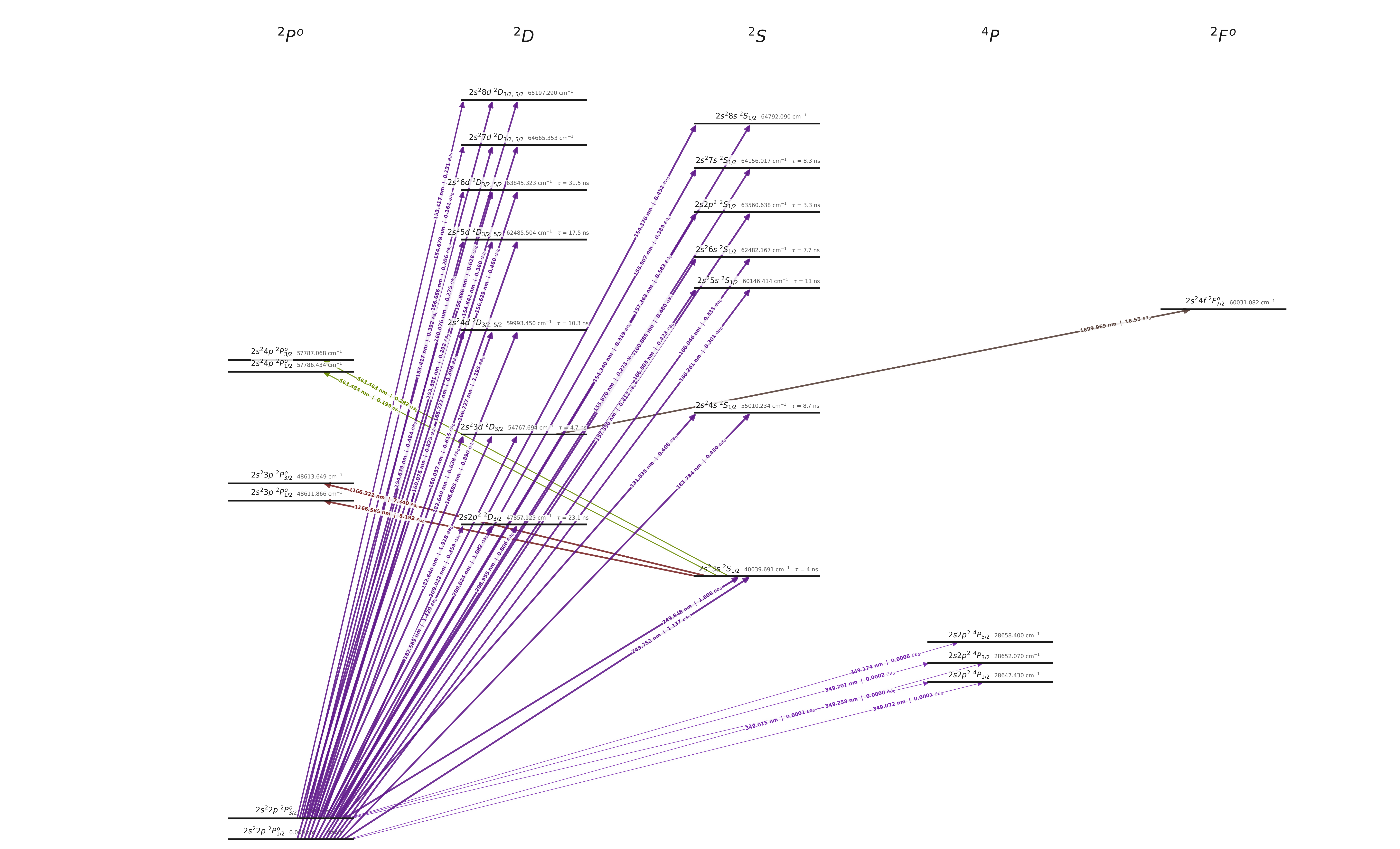


Boron

B (B I). Strongest classified transitions below 2 μm (NIST): 31 levels, 45 lines.

B⁺ ionisation limit 66928.04 cm⁻¹ = 8.29802 eV (149.414 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er U' (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2s ² 2p 2p _{1/2} ^o - 2s ² 3s 2s _{1/2}	249.7523	249.6770	1200.359034	1.137	8.400e+07	39.79	33.6	249.75 nm
2s ² 2p 2p _{3/2} ^o - 2s ² 3s 2s _{1/2}	249.8476	249.7723	1199.9014	1.608	1.680e+08	39.79	67.2	249.85 nm

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹¹B hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2s ² 2p 2p _{1/2} ^o	366.0765(15)	-	F=1: -457.60 F=2: +274.56
2s ² 2p 2p _{3/2} ^o	73.3496(4)	2.6927(10)	F=0: -271.69 F=1: -201.04 F=2: -57.03 F=3: +165.71
2s ² 3s 2s _{1/2}	238.0(4)	-	F=1: -297.50 F=2: +178.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
2s ² 2p 2p _{1/2} ^o - 2s ² 3s 2s _{1/2}	249.752	10-11	5031.3(20)

Shift = ν(first isotope) – ν(second isotope).